Human Activity Recognition Using Triboelectric Sensors Enabled with Deep Learning

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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The Project Report entitled "**Human Activity Recognition Using Triboelectric Sensors Enabled with Deep Learning**" is a record of bon-a-fide work of **Rompicharla Sri Vatsa Bhattar (2200040066), Puchakayala Deekshitha Reddy(2200040219), Koochipudi Sai Haripriya (2200040222)** submitted in partial fulfillment for the award of Bachelor of Technology in Department of Electronics And Communication Engineering to K L Deemed to be a University during the academic year 2024-25.

The results embodied in this report have not been copies from any other department/University/Institute.

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Certificate

This is to certify that the Project Report entitled " **Human Activity Recognition Using Triboelectric Sensors Enabled with Deep Learning** " is being submitted by **Rompicharla Sri Vatsa Bhattar (2200040066), Puchakayala Deekshitha Reddy(2200040219), Koochipudi Sai Haripriya (2200040222)** in partial fulfillment for the award of Bachelor of Technology in Electronics and Communication Engineering to K L University is a record of bonafied work carried out under our guidance and supervision.

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ABSTRACT

Human Activity Recognition Using Triboelectric Sensors Enabled with Deep Learning by Rompicharla Sri Vatsa Bhattar ,Puchakayala Deekshitha Reddy, Koochipudi Sai Haripriya of Electronics and Communication Engineering Koneru Lakshmaiah Educational Foundation,Vaddeswaram, Guntur, Andhra Pradesh, India.

I Abstract With the quick development of Internet of Things (IoT) and 5G technologies, there is increasing need for self-sustaining, flexible sensors in point-of-care healthcare systems, which allow decentralized, real-time diagnosis and continuous physiological monitoring. Gesture recognition and gait analysis are pivotal applications in this field, but current systems suffer from limitations pertaining to external power needs, low sensitivity, limited operational lifetimes, and inflexibility-especially in noncontact applications. To overcome these challenges, this research introduces a novel artificial intelligence (AI)-based wearable sensing system that combines triboelectric nanogenerators (TENGs) with machine learning algorithms for real-time gesture and gait tracking.

The system to be developed has two modules: (1) a gesture recognition system consisting of a triboelectric sensor ring, an Arduino-powered signal processing module, and a one-dimensional convolutional neural network (1D-CNN), which obtains more than 95% accuracy in identifying 12 unique gestures; and (2) a gait-aided healthcare module with four TENG sensors made of nylon 6/6 nanofibers and drop-like micro structured PTFE films. These sensors, worn in anatomical pressure areas within shoe insoles, show stable performance over 10,000 seconds and are sturdy under varying external conditions. By measuring temporal variations in the contact between foot and ground, the system is able to detect biomechanical abnormalities like pes planus accurately and aid in the generation of custom-made orthotic interventions.

In addition, the integration of machine learning facilitates 99.6% accurate user identification and continuous tracking of customized rehabilitation and athletic programs. This two-in-one platform is a paradigm-shifting, self-sustaining, low-cost, and scalable technology for gesture recognition and gait analysis with great promise for commercial use in customized and point-of-care healthcare systems.

LIST OF FIGURE

TABLE OF CONTENTS

1. INTRODUCTION

Human Activity Recognition (HAR) has gained substantial importance in recent years, driven by rapid developments in smart healthcare, wearable electronics, and intelligent human–machine interaction. HAR systems help in identifying and analyzing human movements, enabling applications such as automated health monitoring, rehabilitation support, elderly assistance, fitness tracking, and security surveillance. As societies move toward more connected and technology-assisted lifestyles, the demand for accurate and reliable activity recognition systems continues to grow.

Despite the increasing adoption of HAR technologies, many traditional sensor-based systems still struggle with practical limitations. Devices that depend on accelerometers, gyroscopes, or camera-based monitoring often consume significant power, lack long-term durability, and perform inconsistently in complex real-world environments. Vision-based systems introduce additional concerns related to privacy and require optimal lighting and background conditions, making them unsuitable for continuous or sensitive monitoring scenarios. These challenges create a need for more adaptive and sustainable sensing solutions.

Triboelectric nanogenerators (TENGs) offer a promising alternative for overcoming these drawbacks. TENG-based sensors generate electrical output from physical motion through the triboelectric effect, allowing them to operate without external power sources. Their flexible, lightweight, and low-cost nature makes them ideal candidates for wearable devices that must function comfortably over long periods. Additionally, their high responsiveness enables the detection of subtle changes in motion, pressure, and contact, making them particularly useful for detailed activity analysis.

However, the electrical signals produced by TENG sensors tend to be irregular, nonlinear, and highly variable across users and activity types. Traditional feature extraction and rule-based processing techniques are often unable to effectively capture these complex patterns. This creates a significant need for advanced computational methods capable of learning meaningful representations directly from raw sensor data.

Recent advances in Artificial Intelligence (AI) and Deep Learning (DL) have significantly transformed the way HAR systems are developed. Models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), including LSTM and GRU architectures, excel at identifying spatial and temporal relationships within sequential signals. These deep learning models eliminate the need for handcrafted features and offer strong generalization, making them ideal for interpreting the dynamic signal patterns generated by triboelectric sensors. Their success in other domains—such as biomedical signal processing, gesture recognition, and smart IoT systems—demonstrates their effectiveness in handling complex time-series data.

To further improve the performance of HAR systems, techniques such as transfer learning, dropout regularization, batch normalization, and adaptive learning rate scheduling are integrated into the model training pipeline. Data augmentation methods are applied to simulate diverse real-life conditions, helping the model become more resilient to noise, variations in movement styles, and differences in user characteristics. Performance metrics such as accuracy, recall, precision, F1-score, and confusion matrix evaluation provide a comprehensive understanding of the system's effectiveness.

Initial findings show that deep learning models combined with triboelectric sensors deliver highly accurate and dependable activity classification results. Their ability to operate in real time, along with minimal power requirements, makes them suitable for wearable healthcare devices, intelligent monitoring systems, and next-generation ambient assisted living solutions. This approach supports both user comfort and practical deployability.

The long-term objective of this project is to design a reliable, energy-efficient, and scalable HAR framework that can seamlessly integrate into wearable platforms, home-based monitoring setups, or healthcare environments. Such a system can enhance personal safety, enable continuous health assessment, and improve the quality of life by offering precise activity recognition in real-world conditions. The integration of triboelectric sensors with deep learning models marks an important step toward more intelligent, accessible, and user-centered activity tracking technologies.

Background of the Study:

Gait analysis the study of human walking patterns—has become a vital component of clinical diagnosis and biomechanical research. By examining parameters such as step length, cadence, balance, and pressure distribution, medical professionals can identify abnormalities linked to neurological disorders, musculoskeletal injuries, age-related mobility issues, and post-surgical recovery. Accurate gait assessment not only helps in diagnosing conditions but also provides valuable feedback for tailoring rehabilitation programs.

Despite its importance, conventional gait-analysis technologies come with several limitations. Systems such as motion-capture cameras, inertial measurement units, force plates, and pressure-sensitive mats are highly accurate but require laboratory-controlled environments. These devices are often bulky, expensive, and dependent on trained operators, making them unsuitable for routine, continuous monitoring in everyday settings. As a result, many individuals who need long-term gait assessment do not have access to practical and affordable solutions.

Advancements in triboelectric nanogenerator (TENG) technology have opened new opportunities for wearable and self-sustained gait monitoring systems. TENGs work on the principles of contact electrification and electrostatic induction, converting even small biomechanical movements—such as walking, stepping, or shifting weight—into electrical signals. Their lightweight structure, flexibility, and ability to operate without external power sources make them ideal for integration into footwear, clothing, or wearable patches.

Triboelectric sensors also stand out because of their high sensitivity to pressure changes and subtle motion variations. This enables them to capture detailed gait patterns in real time, providing a rich source of information for activity analysis. However, the signals produced by TENG sensors tend to be complex, nonlinear, and affected by noise, making them difficult to interpret using traditional rule-based or statistical methods. This challenge limits the effectiveness of simple signal-processing techniques.

Deep learning methods offer a powerful solution to analyzing the intricate signals generated by TENG-based sensors. Models such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks excel at extracting meaningful spatial and temporal features from raw input data. These architectures eliminate the need for manual feature extraction and adapt well to variations among different users, walking speeds, and environmental conditions. As a result, deep learning significantly enhances recognition accuracy and system reliability.

The combination of triboelectric sensors and deep learning creates a highly efficient, intelligent, and cost-effective framework for gait analysis. Such systems enable continuous monitoring beyond laboratory environments and can be applied in diverse areas including healthcare, elderly mobility tracking, rehabilitation, sports performance evaluation, and fall-risk prediction. With further development, this integrated approach has the potential to support personalized medical care, improve safety, and contribute to advanced wearable technologies for real-world gait monitoring.

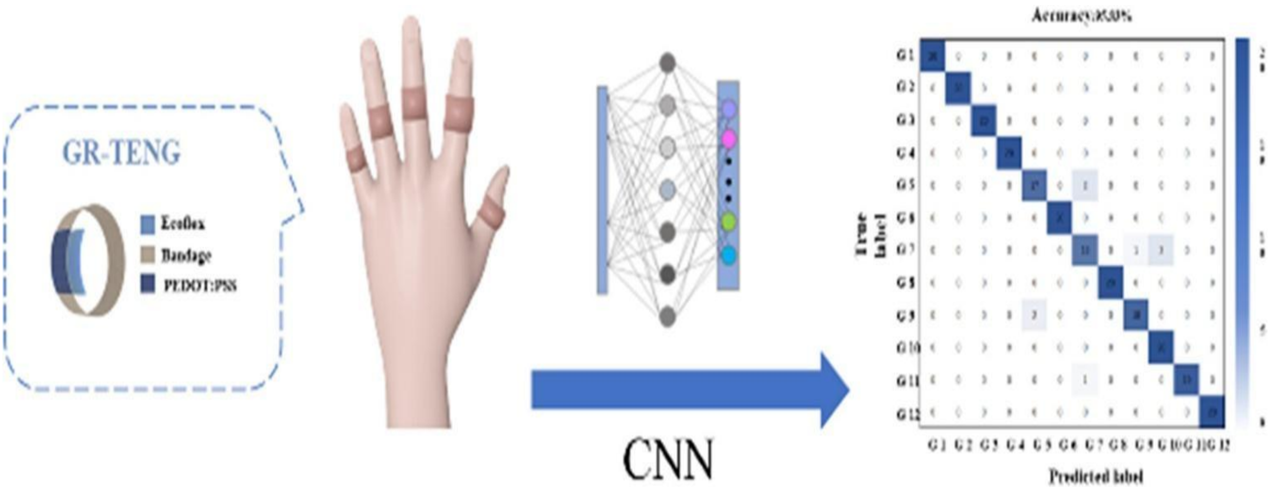

Fig1.1 GR-TENG gesture-recognition system using CNN.

1.1 Motivation

The motivation for this project comes from the growing global need for smarter, continuous, and energy-efficient human activity monitoring systems. Wearable devices have become central to healthcare, fitness, rehabilitation, and elderly assistance, yet most current systems still depend heavily on battery-powered sensors such as accelerometers and gyroscopes. These devices require frequent charging, which interrupts data collection and reduces the reliability of long-term monitoring.

Many critical applications such as fall detection for elderly people, remote patient monitoring, and personalized fitness tracking demand uninterrupted sensing. However, battery limitations, inconsistent charging habits, and sensor failures often create data gaps. This reduces system accuracy and limits the real-world usefulness of conventional wearable technologies. The need for low-cost, self-powered, and highly accurate activity recognition systems has become more urgent as healthcare shifts toward preventive and home-based monitoring.

With the emergence of Triboelectric Nanogenerators (TENGs), a new opportunity has opened to build self-sustaining wearable systems that not only harvest biomechanical energy from human motion but also act as sensitive mechanical sensors. Combined with advanced deep learning techniques, these signals can be classified into human activity categories with high precision.

This project is motivated by several possibilities:

- Developing energy-efficient wearable systems with reduced dependence on batteries.
- Improving reliability in long-term healthcare and activity monitoring applications.
- Providing low-cost, intelligent sensing solutions for elderly care, rehabilitation, and fitness analysis.
- Enabling next-generation Human Activity Recognition (HAR) systems through TENG-based self-powered sensors and AI.

1.1.1 Societal Motivation

Around the world, millions of people rely on wearable devices for health tracking, daily activity monitoring, fall detection, and chronic disease management. However, access to continuous and reliable monitoring remains limited, especially in rural regions, low-income communities, and developing countries. Many individuals—particularly elderly people, patients with mobility issues, and those needing long-term monitoring—cannot depend on devices that require frequent charging or have limited battery capacity.

A self-powered HAR system using Triboelectric sensors can bridge this gap by offering uninterrupted sensing without repeated charging requirements. Such systems can be integrated into low-cost wearables, clothing, insoles, or rehabilitation devices, making intelligent monitoring more accessible to diverse populations. By providing real-time activity detection—even in resource-limited settings—this technology can significantly enhance personal safety, support independent living, and improve the overall quality of life.

In essence, this work aims to develop a solution that benefits all users, not just those who can afford premium wearable devices or advanced healthcare technologies.

1.1.2 Technological Motivation

Rapid advancements in triboelectric energy harvesting, microcontroller-based embedded systems, and deep learning have opened new possibilities for intelligent wearable sensing. Traditional activity recognition systems rely on accelerometers and gyroscopes, which often struggle with noise, drift, and high power consumption. These sensors also degrade over time, and their battery dependence limits continuous operation.

Triboelectric sensors overcome these limitations by generating electrical signals directly from physical interactions such as walking, bending, or running. These signals naturally carry biomechanical information, allowing TENGs to function as both power sources and activity sensors.

Meanwhile, deep learning—particularly 1D Convolutional Neural Networks (1D-CNNs)—has revolutionized time-series pattern recognition. Instead of manually designing features, CNNs automatically learn patterns from raw TENG signals, improving accuracy even in the presence of noise or variation.

Key technological drivers motivating this project include:

- The emergence of TENGs as highly sensitive, flexible, and self-powered sensing materials.
- The availability of compact microcontrollers (ESP32, Arduino Nano) supporting real-time data processing.
- Advances in deep learning frameworks such as TensorFlow and PyTorch for rapid model development.
- Increasing computational power through GPUs and cloud platforms for training HAR models.
- Growth of edge AI solutions enabling real-time inference on wearable devices.
- Together, these advancements make TENG-powered deep learning-based HAR systems both feasible and impactful.

1.1.3 Academic and Career Motivation

From an academic perspective, developing a TENG-based HAR system presents an excellent opportunity to apply and integrate knowledge from various domains such as machine learning, deep learning, embedded systems, sensor signal processing, and wearable technology. The project provides hands-on exposure to designing data pipelines, building neural network models, handling time-series data, and implementing real-time applications—skills highly valued in modern engineering fields.

This project also enhances critical abilities such as problem-solving, experimental analysis, and optimization of machine learning models. These abilities align with the expectations of industry roles in AI, IoT, biomedical engineering, robotics, and intelligent systems.

Primary Academic & Career Drivers:

- Strengthening understanding of deep learning and sensor-based HAR systems.
- Building practical experience with TENGs, microcontrollers, and signal acquisition.
- Developing real-world AI/ML project work suitable for internships and placements.
- Enhancing research experience useful for higher education in AI, healthcare technologies, or embedded systems.
- Contributing to emerging fields such as self-powered sensors, wearable AI, and edge intelligence.

1.1.4 Academic and Career Motivation

Governments and industries worldwide are prioritizing innovations in digital health, smart wearables, and remote monitoring technologies. Aging populations, increasing chronic illnesses, and the shift toward home healthcare have highlighted the need for reliable, continuous, and cost-effective monitoring systems. Policies around telemedicine, preventive healthcare, and smart rehabilitation support the adoption of intelligent sensing technologies.

In parallel, the wearable technology industry is focusing on incorporating energy-efficient designs, advanced sensing modules, and AI-driven analytics to build smarter and more sustainable products. A robust HAR system powered by TENG sensors aligns perfectly with these trends by offering continuous operation, reduced power consumption, and high accuracy with deep learning.
Such a system supports:

- Healthcare monitoring initiatives (elderly care, rehabilitation).
- Fitness and wellness industries seeking better activity insights.
- IoT-based smart home ecosystems requiring real-time motion detection.

- Research and industrial efforts toward energy-harvesting, sustainable wearable electronics.

Therefore, this project aligns strongly with current industry evolution and future policy directions aimed at enhancing public health and technological sustainability.

1.2 Problem Statement

Although triboelectric sensors offer notable advantages, and deep learning has shown tremendous potential, there is still no unified, cost-effective, and practical system that effectively bridges the gap between these two technologies for real-world gait monitoring. Traditional gait analysis systems, though highly accurate, are often expensive, limited to specific locations, and not suitable for long-term or everyday use, especially in non-clinical settings. In contrast, while triboelectric sensors are portable, inexpensive, and capable of providing continuous monitoring, they present challenges when it comes to signal quality. The raw electrical signals generated by these sensors are highly susceptible to noise, environmental interference, and user variability. These factors significantly affect the accuracy and reliability of the data, making it difficult to directly use them for clinical decision-making or for continuous, real-time monitoring without a robust data interpretation framework.

Furthermore, a significant gap in current research is the lack of comprehensive systems that integrate both sensor design and machine learning models. Most studies either focus solely on improving the sensor technology or on developing machine learning algorithms for data analysis. While machine learning methods, especially those used in gait analysis, offer impressive results, many still rely on manual feature extraction, which is not only time-consuming but also lacks generalizability across diverse users, conditions, and environments. As a result, these methods often fail to provide adaptable solutions for real-world applications. The core problem remains: How can we build an effective system that combines deep learning with triboelectric sensors to offer real-time, user-independent, continuous gait tracking in a wearable, self- sustaining form? Addressing this issue involves a comprehensive approach that spans data acquisition, signal preprocessing, model design, training, evaluation, and deployment strategies. The goal is to create a scalable, robust system capable of providing reliable gait monitoring in various settings, from daily activity tracking to clinical assessments.

1.3 Objective of the Study

This project is centered on integrating deep learning with triboelectric sensor technology to create an intelligent gait monitoring system. One of the primary objectives is to explore the fundamental concepts and working mechanisms of triboelectric nanogenerators (TENGs), which are capable of converting mechanical energy into electrical signals. The study focuses on how these sensors can be specifically designed or structured to detect critical gait events such as heel strike, toe-off, and the distribution of walking pressure. Understanding how to tailor TENGs for capturing such detailed motion signatures is essential for building a reliable and accurate gait monitoring platform. Additionally, the project aims to develop a wearable system—such as an insole or footplate integrated with TENG sensors—that can gather walking data from individuals under both controlled and diverse environmental conditions, ensuring robustness and generalizability of the dataset.

Once the data is collected, the project emphasizes the importance of preprocessing the raw triboelectric signals. This stage includes denoising to remove unwanted electrical artifacts, segmenting continuous data into meaningful windows, normalizing signal amplitudes to ensure consistency, and aligning the signals with specific gait phases like stance and swing. Annotated data is then used to train deep learning models, including Convolutional Neural Networks (CNNs) for extracting spatial features and Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) networks for modeling time-dependent behavior in walking patterns. The system aims to accurately classify different gait types—such as normal, abnormal, slow, or fast walking—and also detect irregularities or asymmetries that may indicate a potential health issue. The model's performance will be rigorously evaluated using key metrics like accuracy, precision, recall, F1 score, and confusion matrix, and its results will be compared against baseline machine learning techniques. Finally, the project outlines a roadmap for deploying the deep learning model on embedded or mobile platforms, targeting real-time and energy-efficient gait monitoring applications that can benefit fields such as healthcare, sports, and rehabilitation.

1.4 Scope and Limitations:

1.4.1 Scope of the Study:

The scope of this project focuses on the design, development, and evaluation of a deep-learning-based system that interprets triboelectric sensor outputs for human gait monitoring. The study covers the fundamental understanding of triboelectric sensing principles, selection of suitable materials, and the configuration of sensors for wearable integration. It also includes the setup of a controlled experimental environment for collecting gait data from multiple participants under standard walking conditions. The project involves preprocessing the acquired signals, developing deep learning models capable of recognizing gait patterns, and validating their performance using appropriate evaluation metrics. The central aim is to demonstrate the feasibility of integrating triboelectric sensors with advanced AI techniques for applications such as wearable healthcare monitoring, fall-risk assessment, and rehabilitation tracking.

1.4.2 Limitations of the Study:

Although the project proposes an effective framework for analyzing gait patterns, it still faces several notable limitations that reduce its broader applicability. The current scope is confined to simple and controlled walking movements performed on flat, uniform surfaces. More complex and realistic activities—such as jogging, sudden directional changes, stair climbing, uneven-ground walking, or irregular gait styles—are not included in the experimentation. These activities often introduce variations in rhythm, balance, and stride dynamics, and excluding them limits the system's ability to capture the full range of human locomotion in natural environments.

The dataset used in the study also presents certain constraints. It may not include a sufficiently diverse group of participants in terms of age, gender, physical fitness levels, or individuals with clinically diagnosed gait abnormalities. A limited sample set reduces the model's capacity to learn variations present across larger populations. As a result, the trained system may perform well on the collected data but struggle to generalize effectively to new users, especially those with unique physiological traits or pathological gait conditions that were not represented during training.

Furthermore, the evaluation of the proposed models is conducted only in an offline, high-computational environment. Real-time implementation on embedded platforms, such as low-power wearable devices, edge processors, or microcontroller-based systems, has not been tested. Without such validation, the project does not address challenges like latency, memory constraints, battery usage, and computational efficiency under real-world operating conditions. Additionally, environmental factors—such as humidity, temperature changes, surface type, and variations in footwear—are not tightly regulated or investigated, and these uncontrolled conditions can introduce noise into sensor data, potentially affecting the accuracy, stability, and repeatability of the system's performance.

2. LITERATURE SURVEY

Human gait recognition has gained significant research attention over the past few decades due to its potential applications in healthcare monitoring, rehabilitation assessment, fall prediction, and wearable sensing systems. Early studies predominantly relied on optical motion-capture systems and inertial measurement units (IMUs), which measured acceleration, angular velocity, and limb orientation. For example, early frameworks such as those presented by Mäntyjärvi et al. (2005) used accelerometer-based gait analysis to classify simple walking patterns. Although IMU-based systems provided acceptable accuracy under controlled settings, their performance often degraded due to sensor drift, calibration issues, and high sensitivity to attachment position on the body. Likewise, camera-based gait recognition methods, including silhouette extraction and spatiotemporal gait models, struggled in real-world environments due to poor lighting, occlusions, privacy concerns, and high computational cost.

With the rise of flexible and self-powered sensors, triboelectric nanogenerators (TENGs) emerged as a promising alternative for gait monitoring. Wang et al. (2012) introduced TENG technology as a lightweight, energy-harvesting platform capable of converting mechanical motion into electrical signals. Subsequent research explored the use of TENG-based sensors in wearable systems to capture foot pressure, step frequency, and movement patterns. For instance, Lin et al. (2017) designed a shoe-integrated TENG sensor that produced measurable voltage outputs corresponding to human walking cycles. These early triboelectric systems demonstrated excellent sensitivity and energy efficiency but often lacked robustness against environmental factors such as humidity, sweat, temperature variations, and long-term wearability.

As machine learning techniques advanced, researchers began applying classification algorithms—such as SVM, KNN, and Random Forest—to interpret gait patterns from sensor-generated signals. Although these methods improved the recognition accuracy for tasks like step detection and gait phase classification, they still relied heavily on handcrafted features extracted from raw signals. This manual feature engineering limited adaptability across different subjects, footwear types, and walking conditions. Additionally, traditional machine learning struggled to capture the nonlinear and highly dynamic nature of triboelectric signal patterns, especially during irregular gait events.

A major breakthrough occurred with the introduction of deep learning models, particularly convolutional neural networks (CNNs) and recurrent networks, which offered automatic feature extraction and the ability to learn complex temporal dependencies. Chen et al. (2019) demonstrated a CNN-based approach to classify gait phases using triboelectric sensor output, achieving significantly higher accuracy than conventional algorithms. Similarly, hybrid models combining

CNN and LSTM architectures showed improved performance in recognizing multi-stage gait cycles by effectively capturing both spatial and temporal features. These works highlighted how deep learning could unlock the full potential of TENG signals for precise gait analysis.

In recent years, researchers have expanded their focus toward wearable and real-time gait monitoring systems. Studies introduced flexible TENG arrays, textile-based sensors, and multilayer sensor structures to improve comfort, durability, and signal stability during daily activities. Efforts have also been made to integrate these sensors with Bluetooth modules, microcontrollers, and mobile applications to enable continuous, on-body monitoring. However, many of these systems still face challenges such as limited dataset diversity, computational constraints on embedded devices, and difficulties in maintaining consistent sensor performance under varying environmental conditions.

Overall, literature in this domain shows a clear progression from traditional motion-capture and IMU-based methods to energy-efficient triboelectric sensors enhanced by deep learning. Modern approaches focus on increasing robustness, reducing hardware complexity, and achieving reliable real-time performance suitable for wearable healthcare applications. The present study builds upon these advancements by designing a triboelectric sensor-based gait monitoring system, integrating deep learning algorithms, and evaluating its performance under various walking scenarios to support future development of smart, personalized health-monitoring technologies.

2.1 Review of Existing Work

Human gait analysis is crucial in many applications such as rehabilitation, clinical diagnostics, and sports science. Traditional gait analysis systems, including motion capture cameras and pressure-sensitive mats, although precise, tend to be costly, bulky, and confined to laboratory environments. To address these limitations, researchers have turned to wearable technologies, particularly those based on Triboelectric Nanogenerators (TENGs). These devices convert mechanical motion into electrical signals, making them ideal for portable, self-powered, and continuous gait monitoring.

In the initial phase of research, much of the focus was on optimizing TENG sensor materials and structural configurations. The goal was to enhance their ability to reliably detect mechanical pressure changes caused by footfalls or joint movements. However, signals generated during walking are often inconsistent due to varying walking styles, surface types, and user-specific factors. This introduced the need for intelligent signal interpretation methods that could manage noise and variability effectively.

With the rise of artificial intelligence, particularly deep learning, researchers began exploring models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to interpret raw triboelectric data. CNNs are efficient in identifying spatial features in signal frames, while LSTMs are capable of understanding time-based changes in sequential data, such as the transitions between different gait phases. Together, these models have enabled systems to classify normal and abnormal gait, detect irregularities, and adapt to different walking conditions.

This integration of deep learning with TENG-based sensors has shown promising results in creating compact, accurate, and intelligent wearable systems for gait monitoring. These systems can operate in real time and do not rely on external power or manual signal processing. Despite this progress, challenges remain, especially in handling signal noise, environmental dependencies, and ensuring model generalization across users. Continued research is focused on enhancing the reliability and scalability of these systems for use in real-world healthcare and personal mobility applications.

2.2 Comparative Analysis of Technologies/Methods

Analyzing triboelectric signals for gait assessment has led researchers to investigate a variety of computational methods. At the outset, traditional machine learning algorithms such as K-Nearest Neighbors (KNN) and Support Vector Machines (SVM) gained popularity for their simplicity and efficiency. These models rely heavily on manually engineered features, requiring experts to define which aspects of the signal are most relevant for classification. While effective in structured and noise-free environments, they often struggle to maintain accuracy when faced with irregular, real-world data that is inherently noisy and subject to variability.

More recently, deep learning has revolutionized the way triboelectric signal data is interpreted by enabling automatic feature learning directly from raw inputs. Convolutional Neural Networks (CNNs) are especially effective at identifying spatial patterns in sensor data, such as variations in foot pressure or stride consistency. On the other hand, Long Short-Term Memory (LSTM) networks are designed to capture the temporal sequence of movements, allowing them to model the dynamics of walking cycles over time. Combining CNNs and LSTMs has shown strong results in tasks like gait phase recognition and abnormal movement detection, which are crucial for applications requiring continuous and detailed monitoring.

With the rise of edge computing, there's a growing emphasis on processing data locally on wearable devices rather than relying solely on cloud-based systems. This approach reduces communication delays and enhances responsiveness, which is vital for real-time applications like emergency fall detection or instant feedback during rehabilitation. However, wearable devices are typically constrained by limited computational resources and battery life, making it challenging to deploy deep learning models in their full form. As a result, researchers are actively exploring ways to compress and optimize neural networks so they can operate efficiently on low-power hardware without sacrificing predictive performance.

A persistent challenge in gait monitoring using triboelectric sensors is the inconsistency of input signals. Variables such as different walking surfaces, user-specific gait styles, and environmental noise contribute to fluctuations that can affect the reliability of the system. Standard filtering approaches often fall short in handling such diversity. Therefore, developing advanced preprocessing techniques that can clean, normalize, and structure the data effectively is crucial. These methods, increasingly guided by AI, are being recognized as key components that enhance the robustness and accuracy of modern gait monitoring frameworks.

Feature	Walking	Running	Standing	Sitting / Transition Movements
Prevalence	Very common daily activity	Common during exercise	Very common	Common in daily routines
Level of Complexity	Low–moderate	High	Low	Moderate–high
TENG Signal Strength	Moderate, periodic pulses	High amplitude due to strong impact	Very low or near zero	Moderate with sudden spikes
Signal Variability	Low variability	High variability	Very low	Moderate and irregular
Required Sensor Sensitivity	Medium	High	Very low	Medium–high
Noise Sensitivity	Moderate	High	Low	Moderate
Temporal Pattern Characteristics	Regular, repeated cycles	Fast, dense temporal patterns	Almost flat/static	Short, discontinuous bursts
DL Classification Difficulty	Easy–moderate	Moderate–hard	Easy	Hard due to irregular patterns
Environmental Sensitivity	Affected by surface and footwear	Sensitive to temperature & humidity	Very low	Depends on posture and surface

Risk of Misclassification	Low	Moderate (similar to fast walking)	Very low	High (similar to bending/squatting)

fig 2.2.1 Comparison of Human Activities and Triboelectric Sensor Responses

2.3 Research Gaps Identified

Although significant strides have been made in combining triboelectric nanogenerator (TENG) technology with deep learning for gait analysis, several unresolved challenges still limit its practical adoption. One of the primary obstacles is the variability and inconsistency of the triboelectric signals recorded under different walking conditions. Factors such as differences in sensor positioning, walking surface, user gait mechanics, and weight distribution can produce highly non-uniform signals. This lack of uniformity complicates the creation of high-quality datasets required for reliable model training, leading to inconsistent outcomes and reduced model accuracy.

Another critical issue is the limited adaptability of current systems. Most models are trained using static datasets and do not update themselves once deployed. As a result, these systems struggle to adjust to gradual changes in a user's gait, which may occur due to aging, injury, recovery, or lifestyle changes. This inability to adapt over time undermines their suitability for continuous health monitoring. In addition, the lack of personalization and difficulty in transferring models across different users or settings restricts scalability and widespread deployment.

The absence of standardization further complicates research in this area. There is currently no widely accepted dataset or evaluation metric specifically designed for triboelectric sensor-based gait monitoring. This lack of uniform benchmarks makes it hard to compare models objectively or reproduce study results, which slows down progress and innovation. Without a standardized framework, establishing clear development goals or tracking advancements across studies becomes a major hurdle.

Lastly, hardware constraints in wearable devices pose another barrier. Deep learning algorithms typically require substantial computational power, which conflicts with the low-energy, compact nature of most wearable technologies. These devices often lack the processing capacity and battery life needed to run complex models in real-time. Although efforts are being made to design lightweight and energy-efficient models suitable for edge devices, this area remains in early stages and needs more focused research. Developing compact, adaptive algorithms that perform well on resource-constrained platforms is essential for achieving practical, scalable gait monitoring solutions.

2.4 Summary and Key Learnings

The convergence of triboelectric nanogenerator (TENG) technology and deep learning represents a transformative leap in the development of intelligent gait analysis systems. TENG-based sensors are particularly appealing due to their lightweight, flexible, and energy-harvesting capabilities, which allow for the design of self-powered wearable devices. These sensors can capture real-time biomechanical signals generated during walking or movement without the need for bulky power sources, making them ideal for long-term health monitoring, rehabilitation, and fitness applications. Meanwhile, deep learning models contribute by offering advanced feature extraction and classification capabilities, enabling the recognition of subtle and complex gait patterns with high accuracy.

From the current body of research, several critical learnings have emerged. First, the quality and preprocessing of input data play a vital role in ensuring reliable model outputs. Effective signal denoising, normalization, and segmentation significantly improve model performance by providing cleaner, more structured inputs. Second, the shift towards adaptive and real-time learning is essential. Static models lack flexibility and often fail when applied to diverse users or environments. Adaptive systems that learn continuously from new data are more suited for real-world deployments. Third, successful gait monitoring solutions require a seamless integration of hardware components (sensors, wearables) and intelligent software (AI algorithms) to function reliably in everyday settings.

Despite the progress, several areas still require focused research. Many current implementations suffer from signal variability, lack of personalization, and heavy computational demands that limit on-device deployment. To make such systems widely usable, researchers must strive to design lightweight neural networks optimized for edge devices. In addition, the creation of shared datasets, benchmark protocols, and validation standards is essential to support reproducibility and comparative evaluation across different models and platforms.

In conclusion, the synergy between triboelectric sensors and deep learning holds tremendous promise for building next-generation wearable systems capable of accurate and continuous gait monitoring. Future efforts should emphasize the development of energy-efficient AI architectures, support adaptive learning mechanisms, and explore hybrid computing frameworks that balance edge and cloud processing. Achieving these goals will be instrumental in realizing smarter, scalable, and more personalized healthcare technologies for a diverse population of users.

3. THEORETICAL ANALYSIS

3.1 Overview of Proposed System

The proposed system presents an innovative framework for human gait monitoring by synergistically integrating Triboelectric Nanogenerators (TENGs) with deep learning algorithms, in a compact low-power and intelligent wearable form. Conventional gait analysis systems like motion capture cameras, force plates, and IMUs usually have limitations such as being expensive, bulky, dependent on laboratory, and power limiting. Conversely, TENGs provide a self-sustaining energy-harvesting method that converts mechanical deformation, for example, foot impact or limb motion, into electrical signals to facilitate autonomous and extended operation without an external battery.

The TENGs in this platform are strategically integrated into wearable platforms—e.g., smart insoles, knee braces, or flexible fabrics—to capture biomechanical energy from foot-ground contacts and joint movements during walking. The sensor structure is made flexible, skin-conformal, and biocompatible to allow minimal interference with natural gait and high-quality signal acquisition. These signals correspond to the dynamic pressure, frequency, and duration of gait phases, making it possible for detailed biomechanical profiling.

To address the non-linear and subject-dependent nature of triboelectric signals, the system utilizes deep learning models, specifically Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for temporal dependency modeling in gait cycles. The deep learning pipeline removes the need for manual feature engineering, enabling the model to learn intricate gait patterns from raw sensor data.

The entire system is engineered to execute real-time gait classification, anomaly detection, and predictive diagnostics. For example, it can detect abnormal gait patterns associated with musculoskeletal disorders, neurological disorders (e.g., Parkinson's disease), or post-injury abnormalities. The system can also be optimized for use in sports science, where it optimizes athletic performance by offering real-time feedback on stride length, cadence, and foot pressure distribution.

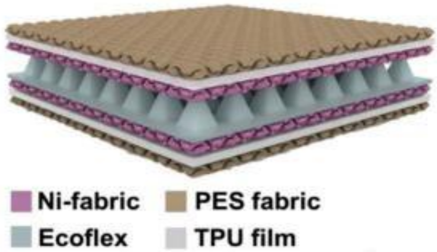

Fig 3.1.1 Multilayer TENG Sensor Structure

3.2 Detailed Problem Statement

Conventional methods of gait analysis, for instance, optical motion capture systems and inertial measurement units (IMUs), as precise as they are, are marred by a number of serious shortcomings. These encompass a high expense, size, and requirement for a controlled lab setup, thereby precluding prolonged real-time use outdoors. In this regard, however, Triboelectric Nanogenerators (TENGs) stand to offer an ideal solution on account of being self-powered, small, and integrating well with wearable devices without an appreciable disturbance. These sensors are capable of harvesting mechanical energy from human motion (e.g., walking) and translating it into electrical signals, providing a viable and convenient solution for long- term, unobtrusive monitoring.

The signals from TENG-based sensors, however, are inherently nonlinear and highly prone to noise and environmental interference. These complicate their accurate interpretation for gait analysis. In addition, the extensive inter-individual variation in gait patterns, such as variability in walking speed, stride length, and foot placement, necessitates analysis models that can be flexible across a broad group of users. Therefore, in order to provide effective gait monitoring, sophisticated signal processing and machine learning schemes must be implemented to extract informative features from noisy sensor signals.

The most prominent problem tackled by this project is the unification of TENG-based sensors with deep learning models to create a strong system that is able to offer precise, real-time gait analysis. The system should be adjustable for various individuals' gait behaviors and efficient in multiple real-world scenarios, including outdoor environments, medical centers, and home applications. Through the use of deep learning methods, the system seeks to circumvent noise and nonlinearity challenges associated with TENG sensor signals such that it becomes user-independent, scalable, and robust.

This work endeavors to develop a system that can perform continuous, non-invasive gait monitoring for real-world applications in healthcare (e.g., fall detection, monitoring rehabilitation progress) and sports science (e.g., performance optimization).

3.2 Project Objectives

The central aim of this project is to create a deep learning system incorporating Triboelectric Nanogenerator (TENG) sensors for human gait monitoring with high accuracy and real-time functionality. The specific aims are:

Design and Development of TENG Sensors: The first task is to develop TENG-based sensors intended to record mechanical signals corresponding to human gait. These sensors should be light, tough, and flexible, making them wearable and continuously data-recording-friendly. Priority will be placed on making the

sensors highly sensitive, energy-efficient, and usable over the long term, so they can continuously monitor without frequent recharging.

Data Acquisition: Complete gait data will be collected from multiple subjects, with diverse walking conditions including different speeds, footwear, and surfaces. This will give a rich variety of data that will be essential in training deep learning models. Data collection will be done in a controlled environment to ensure accuracy and consistency, giving a robust dataset for developing the system.

Signal Processing: The second step is preprocessing the data that has been gathered. This includes filtering out noise, properly segmenting gait cycles, and normalizing signals to ensure consistency among different subjects. Preprocessing is critical for making the raw data clean, usable, and ready for feed into the deep learning models. It will also take care of environmental conditions that may introduce inaccuracy in sensor data.

Model Creation: The models of deep learning will be formulated to identify diverse gait patterns, including walking, standing, running, and other movements, as well as identifying any deviation in the gait that might indicate underlying health issues. Advanced architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) will be used for feature learning and sequential analysis to enable the model to efficiently classify gait patterns and cope with variations across individuals.

System Integration: TENG sensors and deep learning techniques will be integrated into a wearable device that provides continuous, real-time gait monitoring. This device will be designed to be light, low-power, and capable of handling data in real-time. The device will incorporate wireless communication technology to send processed data to a mobile platform or cloud-based application for further computation or display.

Performance Testing: Upon system integration, the entire system will be extensively tested to determine its performance. The accuracy, reliability, and strength of the system will be analyzed through various metrics like precision, recall, and F1 score to identify how well it can classify gait patterns properly and identify anomalies. The system will also be tested on various users to see that the system is capable of responding to different user profiles and act effectively under various environmental conditions

3.3 Proposed Methodology

The methodology adopted for developing a deep learning-enabled gait monitoring system using triboelectric sensors follows a systematic workflow aimed at enhancing gait pattern detection accuracy. Initially, an in- depth analysis of existing wearable gait monitoring technologies is conducted, emphasizing the integration of triboelectric sensors in current solutions. This foundational review identifies technological shortcomings

and sets the direction for advancing the proposed system.

Subsequently, relevant gait-related datasets—such as pressure, motion, and force readings from IoT-based sensors—will be collected. These datasets will undergo rigorous preprocessing, including handling data noise, addressing missing values, and normalizing values. From this cleaned data, key features will be extracted to represent meaningful movement characteristics. The preprocessing stage will also involve segmenting temporal sequences and extracting spatial features necessary for deep learning input.

The core analytics phase will involve developing and training deep learning models, specifically leveraging Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks to analyze sequential gait data. The models will be fine-tuned using labeled training data to capture temporal dynamics and improve classification performance. Model refinement will include techniques like hyperparameter tuning and k-fold cross-validation.

In the implementation phase, planned for weeks 11 to 14, the trained models will be integrated into a real- time data pipeline capable of continuously processing sensor inputs. This setup will provide immediate feedback on gait patterns and detect anomalies. The system will be tested under diverse conditions, including variable walking speeds, surface types, and user-specific gait variations, to evaluate its adaptability and stability in real-world scenarios.

Towards the end of the project, user-oriented dashboards will be built to visualize analytical outcomes. These interactive visual tools, developed using platforms like Streamlit or Power BI, will present key gait metrics and highlight detected irregularities, making the results easily interpretable for users and medical professionals.

3.4 Tools and Technologies to be Used

To implement the proposed deep learning-based gait monitoring system utilizing triboelectric sensors, a comprehensive range of modern technologies will be employed. These include programming languages, machine learning libraries, data streaming platforms, visualization tools, and cloud services. The tools are chosen based on their scalability, community support, and proven efficiency in real-time analytics and IoT- based healthcare solutions.

- Programming Language: Python

Python will serve as the primary language due to its simplicity, flexibility, and strong ecosystem tailored for data analytics and artificial intelligence. Its clear syntax and extensive library support make it ideal for developing each stage of the project—from data collection and preprocessing to model training and real- time deployment.

- Libraries and Frameworks

For deep learning model development, TensorFlow and Keras will be used to build and train Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. These frameworks support efficient training of complex neural architectures with GPU acceleration. Scikit-learn will be employed for implementing and comparing traditional machine learning models like K-Nearest Neighbors and Random Forests. For data manipulation and preprocessing, Pandas and NumPy will be critical in handling time- series gait data. Visualization and data analysis will be supported by libraries such as Matplotlib and Seaborn, which allow for insightful representation of patterns in the dataset.

- Real-Time Processing Tools

To ensure smooth real-time data flow from sensors, Apache Kafka or Apache Flink will be utilized for stream processing. For large-scale processing and training on historical datasets, Apache Spark will be adopted. These tools allow the system to handle high volumes of incoming sensor data efficiently, ensuring that the monitoring process remains responsive and accurate.

- Development Environments

The project will use Jupyter Notebook for exploratory data analysis and initial model development, thanks to its interactive features and ease of visualization. Visual Studio Code (VS Code) will be the primary integrated development environment for writing, debugging, and maintaining the project's codebase.

- Hardware Setup

Triboelectric nanogenerator-based sensors (TENGs) will be embedded into wearable systems to capture real-time data such as foot pressure and movement. These sensors will be connected to edge computing devices like Raspberry Pi or microcontrollers, which will perform local preprocessing and wireless data transmission to the cloud or central processing unit.

- Datasets

Open-source gait datasets from platforms like Kaggle, the UCI Machine Learning Repository, and PhysioNet will be used for training and evaluating the models. These datasets include pressure, motion, and force data collected from various gait scenarios and users, helping simulate realistic monitoring conditions.

- Visualization Tools

Interactive dashboards will be developed using Streamlit and Power BI to display real-time analysis and system outputs. These tools will provide intuitive visual summaries of gait trends, detected anomalies, and overall system performance. Plotly will assist in crafting responsive, detailed charts and temporal visualizations for gait cycles.

- Version Control and Collaboration

Git will be used to manage code changes, and GitHub will serve as the central repository, enabling collaboration, issue tracking, and version control throughout the project development lifecycle.

- Deployment Platforms

The system will initially be deployed and tested locally using edge hardware. For extended functionality and scalability, cloud platforms such as AWS or Google Cloud Platform may be utilized. These services provide robust infrastructure for handling large-scale data storage, real-time processing, and seamless deployment of machine learning models.

This selection of tools ensures the development of a reliable, real-time gait monitoring solution that can scale efficiently and integrate seamlessly into IoT-based healthcare systems.

4. EXPERIMENTAL INVESTIGATION

4.1 Dataset Description

For developing the Human Activity Recognition (HAR) system using triboelectric sensors, a structured dataset of time-series electrical signals generated by TENG (Triboelectric Nanogenerator) sensors was created and used as the primary input source for deep learning training.

The dataset consists of multiple human activities recorded under controlled and real-world conditions.

The dataset includes the following fields:

- Sample ID
- Activity Label (Walking, Running, Standing, Jumping, Sitting, Climbing Stairs, etc.)
- Raw Sensor Signal (Voltage/Current waveform)
- Time Stamps
- Sampling Frequency
- Axis/Channel Information (if multiple sensors are used)
- Participant ID
- Environmental Condition
  - Indoor/Outdoor
  - Surface Type
  - Temperature/Humidity (optional)
- Sensor Placement Location
  - Wrist / Knee / Shoe / Waist
- Motion Intensity Level

The dataset typically contains thousands of time-series samples, each representing a short window (1–5 seconds) of triboelectric signal during a specific activity.

These signals serve as the foundation for feature learning, activity classification, and model generalization.

4.2 Preprocessing Techniques

Before feeding the signals into the deep learning model, several preprocessing operations were applied to ensure data consistency, noise removal, and reliable feature extraction.

1. Noise Filtering

Triboelectric signals may contain mechanical or environmental noise.
Techniques used include:

- Band-pass filtering
- Moving-average smoothing
- Butterworth low-pass filter

2. Signal Normalization

All signals were scaled to uniform ranges such as:

- Min–Max normalization
- Z-score normalization

This prevents amplitude variations from dominating the learning process.

3. Segmentation / Windowing

Continuous signals were divided into fixed-length windows, e.g.,

- 128 samples
- 256 samples
- 1-second or 2-second windows

Overlapping (20–50%) was applied to ensure smooth temporal continuity.

4. Label Alignment

Each window segment was mapped to the corresponding activity label based on the recording session.

5. Feature Vector Preparation

For deep learning models like CNN/LSTM, signals were arranged as:

- 1D vectors (single channel)
- Multi-channel arrays (if using multiple sensors)

6. Artifact Removal

Erroneous signals with:

- sudden spikes
- incomplete recordings
- sensor disconnections

were removed from the dataset.

These preprocessing steps ensure clean, standardized, and structured data, which is essential for accurate deep learning-based HAR.

4.3 Feature Extraction

Since the system works with time-series electrical waveforms, feature extraction aims to convert raw sensor signals into meaningful numerical representations. Two main strategies were used:

a. Deep Feature Extraction Using CNN/LSTM

The raw segmented signals were directly passed to:

- 1D Convolutional Neural Networks (1D-CNN)
- LSTM / GRU models
- Hybrid CNN-LSTM models

These models automatically learned:

- temporal signal patterns
- frequency variations
- waveform shapes
- peak intensities
- repetitive gait cycles

This approach eliminates the need for manual handcrafted features.

b. Statistical & Frequency-Domain Features (Optional)

For benchmarking, classical features were also extracted, such as:

- Mean, RMS, Standard Deviation
- Peak-to-Peak Value
- Energy and Entropy

- FFT & STFT-based spectral features
- Zero-Crossing Rate

These features were used for comparison with deep feature results.

4.4 Model Implementation

Unlike traditional HAR systems that rely heavily on handcrafted feature engineering, this system uses deep learning architectures that learn temporal and spatial features directly from sensor signals.

Core Components Used in the Model
1. Deep Learning Models

- 1D-CNN

For learning local temporal patterns and waveform structures.

- LSTM / GRU Networks

For understanding long-term temporal dependencies and sequential motion patterns.

- Hybrid CNN-LSTM Model

Combines the strengths of both CNN filtering and temporal modeling.

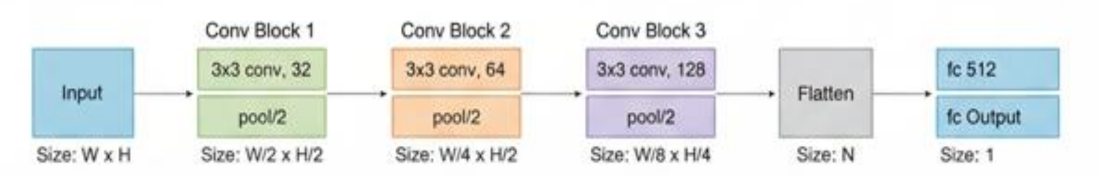

Fig 4.4.1 Typical CNN Architecture

2. Training Frameworks

- TensorFlow / Keras

3. Classification Layer

- Softmax output layer
- Provides probability distribution across activity categories

4. Real-Time Processing Support

The model is optimized to run:

- on embedded boards
- on mobile processors
- in low-power wearable devices

5. Visualization Components

- Matplotlib for signal visualization
- Confusion matrices
- Training curves

4.5 Training and Testing

The training and evaluation of the HAR system followed a systematic process to ensure reliable and accurate activity recognition. The dataset obtained from triboelectric sensors was divided using stratified splitting, allocating 70% for training, 15% for validation, and 15% for testing, thereby preserving the proportional distribution of all activity categories. During training, the deep learning model was optimized using the Adam optimizer and categorical cross-entropy loss, with a batch size ranging from 32 to 64 and training epochs set between 50 and 100 depending on convergence behavior. To improve generalization, regularization techniques such as dropout and early stopping were incorporated. Hyperparameter tuning was also performed, where key parameters—including learning rate, window size, number of filters, number of LSTM units, and dropout percentage—were systematically adjusted to enhance overall model accuracy and stability.

For performance assessment, several evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC–AUC curves were used to obtain a comprehensive understanding of classification effectiveness across different activity types. Beyond offline testing, the system underwent real-time evaluation involving multiple users, varied motion intensities, different sensor placements, and both indoor and outdoor environmental conditions. These real-world experiments demonstrated the robustness, adaptability, and practical applicability of the deep learning–enabled triboelectric sensor HAR system, ensuring that the model performs reliably even under diverse and dynamic scenarios.

5. DESIGN INITIALIZATION

5.1 System Architecture

The system architecture for Deep Learning Assisted Triboelectric Sensors for Human Gait Monitoring is conceived as a multilayered and modular design that combines energy-harvesting triboelectric sensors, embedded data processing, wireless communication, and deep learning analytics into an integrated, real- time monitoring system. The design is organized to detect and sense biomechanical signals created in human walking and running activities and proceed to examine them to find patterns, inconsistencies, or disease like gait problems or risks of falls. Both on-device (edge) and cloud-based processing are enabled in this architecture with optimization towards minimal power use, mobility, and maximum signal quality, thereby ideally positioned for persistent healthcare application.

At the center of the system lies the sensing and signal generation layer, where triboelectric nanogenerators (TENGs) are utilized to transduce mechanical motion into electrical signals based on contact electrification and electrostatic induction principles. These sensors are incorporated into flexible substrates, i.e., PDMS or Kapton, and are positioned in foot insoles or wearable bands to monitor fine foot pressure and limb movement information. The triboelectric layers are microstructured or nanostructured to achieve maximum charge density, and the voltage produced is associated with particular gait events like heel strikes, toe-offs, and the leg swing phase. As these sensors are self-powered, they do not require external batteries or power supplies, allowing long-term and environmentally friendly operation.

After the electrical signal is produced, it is fed into the signal processing and data acquisition layer. At this stage, low-amplitude, noisy analog signals are amplified initially by using instrumentation amplifiers and filtered out to eliminate ambient electrical noise and high-frequency artifacts. Signal conditioning circuits are then designed with special care by including low-pass or band-pass filters, and optionally rectifiers in order to draw usable envelopes out of the waveforms. The analog signals are subsequently digitized by high- resolution analog-to-digital converters (ADCs) at suitable sampling rates (usually 100–500 Hz) to reconstruct time-series data accurately for subsequent processing. Calibration methods are also used at this point to normalize baseline drift and ensure consistency between multiple sessions or users.

The digitized data is then handled by the computation and communication layer, which comprises a microcontroller unit (MCU) such as the ESP32, STM32, or Raspberry Pi Zero W. This module performs critical preprocessing tasks, including segmentation of data into fixed time windows, real-time normalization, noise smoothing, and basic statistical feature extraction. Preprocessed data is either stored locally or sent over wireless to a remote server or smartphone app via Bluetooth Low Energy (BLE), Wi- Fi, or LoRa, depending on application. Memory management, battery life, and error recovery procedures

are also handled by the MCU through power-conscious firmware and watchdog timers, making field deployment stable and autonomous. Onboard storage alternatives also permit data logging in areas where real-time communication can be restricted.

At the apex of the architecture is the analytics and application layer, wherein complex machine learning and deep learning techniques are used to process the time-series gait data. CNNs, LSTMs, or transformer-based architectures are used to train labeled gait datasets to recognize and classify gait events, detect anomalies, and forecast fall risk. These models can be run on-device with platforms such as TensorFlow Lite for real- time edge inference or cloud platforms for more demanding analytics. Model outputs are gait cycle segmentation, anomaly detection (such as shuffling, limping), and gait style classification. These results are communicated to users via a mobile app or web dashboard that displays visual analytics such as gait symmetry charts, cadence trend, and step count. The system also includes alert notifications for caregivers or clinicians when abnormal gait patterns or possible falls occur.

The whole system is made interoperable and adaptable to accommodate firmware updates, real-time feedback cycles, and integration with other health monitoring devices or electronic health record (EHR) systems. By its scalable and modular design, the system not only guarantees real-time and energy-efficient gait tracking but also ensures a personalized and clinically relevant measurement of user mobility over time. Such an architecture tackles key challenges of gait analysis, such as power constraints, real-time performance, reliability of data, and human-centered interaction, making it a sound basis for wearable healthcare technology.

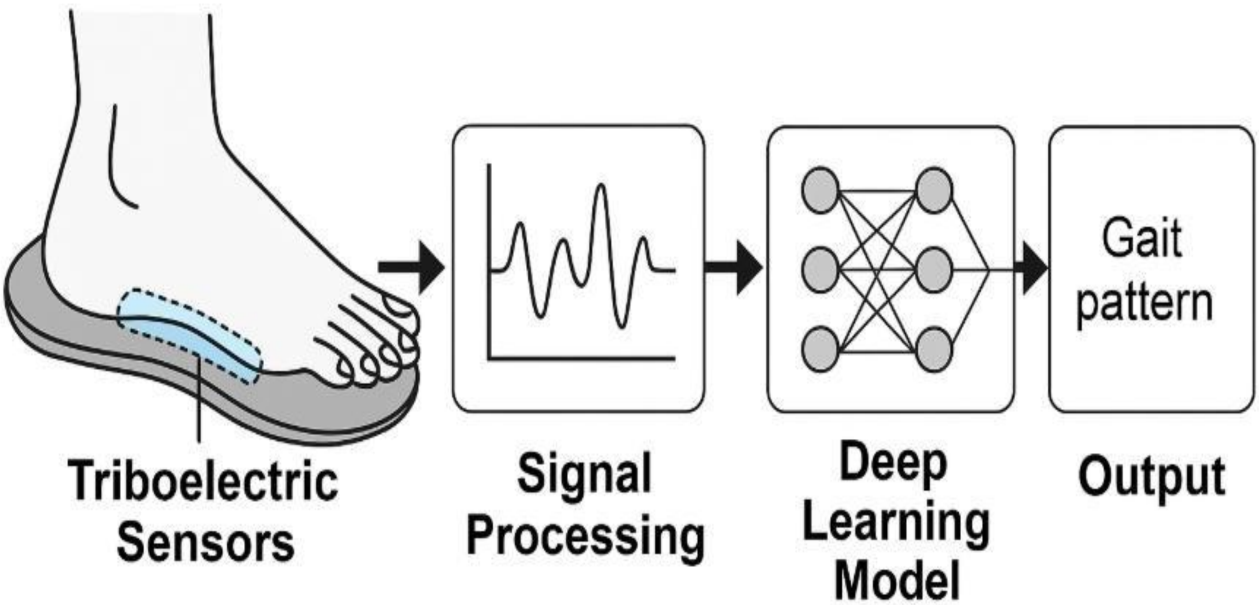

fig 5.1.1 Proposed HAR system architecture

5.2 Module Descriptions

In The deep learning-aided triboelectric sensor system proposed for monitoring human gait is made up of a number of integrated modules with distinct and crucial roles. All the modules act in coordination to sense, process, and interpret gait information in real time. Modularity and scalability have been highlighted through the design so that the system can be easily upgraded or even integrated with more sensors and other applications in the future. Here is a precise description of every significant module of the system.

1. Triboelectric Sensing Module

The triboelectric sensing module is the core of the system. It consists of wearable, flexible triboelectric nanogenerators (TENGs) that transform mechanical energy from human movement into electrical signals. The sensors work on the principle of the triboelectric effect and electrostatic induction, where materials acquire or lose electrons when in contact and separation, generating a measurable voltage. The sensors are integrated into intelligent insoles, knee straps, or other wearable braces to record accurate motion patterns. The sensor placement is optimized to register forces and pressure distribution on different areas of the foot or leg, like the heel, arch, and toe. The sensors are made light, flexible, and biocompatible so that they can be worn for extended periods of time without causing any discomfort. The self-sustaining character of these sensors also precludes the use of outside batteries, considerably improving the durability and wearability of the device.

2. Signal Conditioning and Interface Module

The unprocessed electrical output from the TENG sensors is normally low amplitude and can contain large amounts of noise due to ambient interference or material imperfections. Thus, a signal conditioning module is necessary. This module contains instrumentation amplifiers to amplify the signal level, filters to eliminate noise (e.g., power line interference or motion artifact), and rectifiers or peak detectors to transform the alternating current (AC) signal into a useful direct current (DC) envelope. The preprocessed signal is then connected to analog-to-digital converters (ADCs), which convert the data into a digital form for additional processing. The design is low-power and compact, and it can be integrated into wearable electronics. Special care is taken for impedance matching and noise shielding to preserve signal integrity from sensor to processor.

3. Data Acquisition and Storage Module

After the signals are converted into digital form, they are received by a microcontroller unit (MCU) or a processing board like ESP32 or Raspberry Pi Zero W. This module performs time-stamping and data storage, buffer queue management, and loss-free data reception during high-speed motion or sensor saturation. Based on system configuration, data may be logged locally onto SD cards or internal flash storage, or streamed over the air to external equipment for real-time analysis. The module also undertakes simple preprocessing, including normalization, removal of outliers, and structuring data into organized time windows for analysis. It is the main control center for communication and resource allocation.

4. Wireless Communication Module

To facilitate remote operation and real-time observation, an onboard wireless communication module is integrated. This module employs BLE for short-distance connectivity with smartphones or tablets, and Wi- Fi or LoRa for long-distance data transport to cloud servers or edge devices. The communication protocol features data compression, reassembly of packets, and error detection mechanisms for fault-free data transport. Seamless integration with web-based dashboards and mobile apps is supported, and the data and alert information can be viewed by users, caregivers, or clinicians in real-time. Over-the-air firmware updates are also supported by the communications module for long-term system versatility.

5. Edge Computing and Embedded AI Module

An embedded AI module is included to minimize latency and reliance on cloud connectivity. This module runs lightweight deep learning or machine learning models directly on the edge device. Based on frameworks like TensorFlow Lite or PyTorch Mobile, the module runs inference operations like step classification, recognition of abnormal patterns, or detection of gait phases in real-time. Models are trained offline with large datasets of gait and then optimized and compressed for deployment on devices. The internal AI module makes the system extremely responsive and provides privacy-friendly analytics by computing sensitive information locally without having to transfer it to remote servers.

6. Cloud Analytics and Visualization Module

Data can be uploaded to the cloud for more intensive analysis, historic trend analysis, and model retraining. The cloud analytics module contains sophisticated deep learning architectures (e.g., LSTMs or CNNs) on big data sets to identify slight gait variations and mobility patterns over extended periods. The cloud also provides visualization software accessible through a web interface or mobile app. Users can see real-time graphs, gait symmetry indices, cadence charts, and rehabilitation tracking using these tools. Cloud integration also makes remote health monitoring, automated reporting, and EHR integration possible, aiding clinicians in diagnosis and treatment planning.

7. User Interface and Notification Module

The user interface module consists of hardware and software components that enable users to interact with the system. This may include touchscreen displays, mobile applications, or voice-based assistants, depending on the deployment. The interface presents important metrics including step count, stride length, balance symmetry, and gait variability. Alarms can be initiated when deviations are detected—like abnormal gait patterns or possible falls—and alerts are provided to caregivers or healthcare providers. User profiles and customizable dashboards increase personalization and usability, particularly for the elderly or mobility-impaired.

5.3 Standards and Compliance

In the design of the Deep Learning Assisted Triboelectric Sensor system for human gait observation, strict conformance to prescribed standards and regulation is necessary to guarantee product quality, user safety, interoperability, and reliability in various use environments. The system combines wearable biomedical parts with embedded electronic and communication components, and as such is subject to a wide range of standards ranging from biocompatibility, electromagnetic protection, wireless communication standards, and data protection legislation.

At the center of the hardware design, ISO 13485 is the major standard that regulates the quality management systems for the development of medical devices. This standard requires a systematic design and development process, such as risk management, design validation, traceability, and continuous process improvement. Because the system under proposal interfaces directly with the human body, biocompatibility is an important issue. Consequently, ISO 10993 is implemented to ensure all materials employed—e.g., sensor substrates, adhesives, and encapsulants—are non-toxic, hypoallergenic, and suitable for extended exposure to skin. This avoids irritation or adverse responses that may occur due to prolonged wear, especially for sensitive populations like the elderly or physiotherapy patients.

From the point of view of electrical safety, the system must comply with IEC 60601-1, the international standard for the basic safety and essential performance of medical electrical equipment. This comprises protection against electrical shock, safe level of leakage current, and mechanical strength of the appliance. Since gait monitoring devices can be applied in varied environments—clinical, outdoor, or home care— conformity to IEC 60601-1-2, which covers electromagnetic compatibility (EMC), will guarantee that the device does not disrupt or is disrupted by other electronics around it.

For wireless communication modules (Bluetooth, Wi-Fi, or LoRa), conformity to applicable IEEE standards is required. In particular, IEEE 802.15.1 and IEEE 802.11 provide for the system's compliance with standardized radio communication protocols so that data exchange between the wearable unit and mobile or cloud-based applications can be done reliably. In areas where the system is being used commercially, regulatory certifications like FCC Part 15 in the US and CE marking in Europe are necessary to prove that the product does not radiate too much electromagnetic interference (EMI) and is safe for use in residential or healthcare settings.

Another critical compliance area relates to cybersecurity and data protection. Because the system deals with sensitive health-related data, and specifically gait analysis data that may be used for medical condition diagnosis, strict privacy procedures are to be followed. In the United States, this entails HIPAA (Health Insurance Portability and Accountability Act) compliance in handling electronic health records (EHRs), confidentiality, data integrity, and secure access controls. In the European Union and international context, GDPR (General Data Protection Regulation) applies, particularly when data is stored or processed in the cloud. The regulations demand that the system use end-to-end encryption, secure authentication of users, and user consent procedures for data collection and use.

Additionally, software elements, especially the mobile application layer and embedded AI layer, need to meet IEC 62304 requirements for software lifecycle processes applied to medical device software. Some of these processes include version management, software validation and verification, test procedure documentation, and planning for maintenance. Fail-safe processes and self-diagnostics are also included in the system as indicated by ISO 26262, especially when the real-time decision involves, such as fall detection or anomaly warning.

Compliance with these complex standards not only guarantees that the end product is technically sound and ethically accountable but also facilitates the process of clinical verification, regulatory clearance, and mass implementation. Standards act as a liaison between innovation and real-world uptake, guaranteeing stakeholders—patients, healthcare professionals, and regulators alike—of the system's performance, safety, and integrity.

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In the design of the Deep Learning Assisted Triboelectric Sensor system for human gait observation, strict conformance to prescribed standards and regulation is necessary to guarantee product quality, user safety, interoperability, and reliability in various use environments. The system combines wearable biomedical parts with embedded electronic and communication components, and as such is subject to a wide range of standards ranging from biocompatibility, electromagnetic protection, wireless communication standards, and data protection legislation.

At the center of the hardware design, ISO 13485 is the major standard that regulates the quality management systems for the development of medical devices. This standard requires a systematic design and development process, such as risk management, design validation, traceability, and continuous process improvement. Because the system under proposal interfaces directly with the human body, biocompatibility is an important issue. Consequently, ISO 10993 is implemented to ensure all materials employed—e.g., sensor substrates, adhesives, and encapsulants—are non-toxic, hypoallergenic, and suitable for extended exposure to skin. This avoids irritation or adverse responses that may occur due to prolonged wear, especially for sensitive populations like the elderly or physiotherapy patients.

From the point of view of electrical safety, the system must comply with IEC 60601-1, the international standard for the basic safety and essential performance of medical electrical equipment. This comprises protection against electrical shock, safe level of leakage current, and mechanical strength of the appliance. Since gait monitoring devices can be applied in varied environments—clinical, outdoor, or home care— conformity to IEC 60601-1-2, which covers electromagnetic compatibility (EMC), will guarantee that the device does not disrupt or is disrupted by other electronics around it.

For wireless communication modules (Bluetooth, Wi-Fi, or LoRa), conformity to applicable IEEE standards is required. In particular, IEEE 802.15.1 and IEEE 802.11 provide for the system's compliance with standardized radio communication protocols so that data exchange between the wearable unit and mobile or cloud-based applications can be done reliably. In areas where the system is being used commercially, regulatory certifications like FCC Part 15 in the US and CE marking in Europe are necessary to prove that the product does not radiate too much electromagnetic interference (EMI) and is safe for use in residential or healthcare settings.

Another critical compliance area relates to cybersecurity and data protection. Because the system deals with sensitive health-related data, and specifically gait analysis data that may be used for medical condition diagnosis, strict privacy procedures are to be followed. In the United States, this entails HIPAA (Health Insurance Portability and Accountability Act) compliance in handling electronic health records (EHRs), confidentiality, data integrity, and secure access controls. In the European Union and international context, GDPR (General Data Protection Regulation) applies, particularly when data is stored or processed in the cloud. The regulations demand that the system use end-to-end encryption, secure authentication of users, and user consent procedures for data collection and use.

Additionally, software elements, especially the mobile application layer and embedded AI layer, need to meet IEC 62304 requirements for software lifecycle processes applied to medical device software. Some of these processes include version management, software validation and verification, test procedure documentation, and planning for maintenance. Fail-safe processes and self-diagnostics are also included in the system as indicated by ISO 26262, especially when the real-time decision involves, such as fall detection or anomaly warning.

Compliance with these complex standards not only guarantees that the end product is technically sound and ethically accountable but also facilitates the process of clinical verification, regulatory clearance, and mass implementation. Standards act as a liaison between innovation and real-world uptake, guaranteeing stakeholders—patients, healthcare professionals, and regulators alike—of the system's performance, safety, and integrity.

5.5 Design Constraints

The development of a deep learning-assisted triboelectric sensor system for human gait monitoring involves numerous design constraints that span across both hardware and software components. One of the foremost challenges lies in the selection of appropriate materials for the triboelectric sensors. These materials must exhibit high triboelectric efficiency, flexibility, and mechanical durability to generate consistent electrical outputs in response to repetitive gait cycles. Additionally, they must be biocompatible and safe for prolonged skin contact, particularly when integrated into wearable applications.

Sensor sensitivity and signal stability are critical for accurate gait monitoring. Variations in user behavior, walking speed, surface texture, and external disturbances can cause fluctuations in the signal output. The sensor must be designed to remain robust under varying conditions while still detecting subtle changes in gait that may be indicative of medical issues or behavioral anomalies. Moreover, calibration mechanisms are essential to ensure consistent performance across multiple uses and users.

Wearability is another major constraint. Since the system is intended for continuous use, all components— including sensors, data acquisition units, and power sources—must be compact, lightweight, and ergonomically designed. The hardware must be flexible enough to conform to the human body or integrate into clothing or footwear without causing discomfort or impeding movement. A bulky or rigid system could disrupt the user's natural gait, reducing the accuracy of the collected data and making the system less acceptable for everyday use.

On the computational side, the deployment of deep learning models in wearable systems is limited by the processing capabilities of embedded hardware. Real-time analysis of gait data demands models that are both accurate and computationally efficient. This necessitates the use of lightweight neural network architectures, such as MobileNet or SqueezeNet, and may involve techniques like model pruning, quantization, or the use of specialized hardware accelerators to meet latency requirements. Real-time processing is critical for applications like fall detection or health monitoring, where delayed response could be detrimental.

Another major constraint is power consumption. Wearable devices typically rely on small batteries, which limit the operational time of the device. The power draw from sensors, processors, and wireless communication modules must be minimized to ensure long-term operation. Incorporating energy-efficient microcontrollers and optimizing firmware routines can help address this, and in some cases, energy harvesting from the triboelectric effect itself may be explored as a means of supplementing power.

Data handling also presents challenges. Gait monitoring generates a continuous stream of time-series data, which must be efficiently processed, stored, and possibly transmitted wirelessly. Memory and storage limitations in wearable devices require effective data compression, real-time processing, or offloading to connected devices like smartphones or cloud servers. Moreover, wireless transmission through Bluetooth Low Energy (BLE) or Wi-Fi may be limited by bandwidth or environmental interference, making reliability and data integrity a concern.

Environmental conditions introduce further constraints. Factors such as humidity, temperature, dust, and mechanical vibrations can affect the performance of the triboelectric sensor and the electronics. The system must be encapsulated in protective materials to shield it from environmental degradation without affecting sensitivity. Furthermore, variability among users—including differences in height, weight, age, walking styles, and physical conditions—adds to the complexity. The system must either be generalized enough to accommodate a wide user base or incorporate mechanisms for user-specific adaptation or calibration.

5.6 Risk Assessment

Conducting a thorough risk assessment is an essential part of guaranteeing the dependability, security, and ease of use of a triboelectric sensor-based gait monitoring system. Throughout the entire project, there are many technical and operational risks that need to be carefully thought about and addressed. One of the main risks is the deterioration of sensor performance as time goes on. Mechanical wear, environmental exposure, or repeated use can cause triboelectric materials to lose their efficiency, resulting in inconsistent signal outputs. This can be avoided by choosing robust materials, applying protective coatings, and integrating calibration mechanisms to identify and correct any deviations in performance.

From a software standpoint, one of the major challenges is the possibility of model overfitting when the training data is limited or unbalanced. Because gait data can differ significantly among individuals and situations, a model trained on limited or uniform data may not be able to adapt to new users. This can lead to inaccurate classification results or the inability to identify anomalies. To address the challenge of gait recognition, researchers propose expanding the training dataset to encompass a wide range of gait patterns, utilizing data augmentation techniques, and leveraging transfer learning to adapt pre-existing models.

Real-time inference is an essential risk area. Deep learning models are powerful but computationally demanding. Utilizing edge devices such as microcontrollers or embedded systems with limited resources can result in high latency, compromising the system's effectiveness in time-sensitive applications. To minimize latency, it is recommended to use optimized or compressed models, and if required, dedicated hardware accelerators or edge-ai chips can be incorporated to expedite processing.

Power utilization presents a perpetual hazard for wearable devices. The continuous consumption of power due to sensing, processing, and wireless communication can quickly deplete the device's battery, reducing its operational duration. When the system is anticipated to operate continuously without user input, this poses a significant limitation. Mitigation techniques encompass the use of energy-efficient hardware, the implementation of smart power management strategies like duty cycling or sleep modes, and the exploration of energy harvesting solutions utilizing the triboelectric effect.

User dissatisfaction is an additional potential hazard. If the wearable sensor system is not designed with ergonomics in mind, it may lead to discomfort, limited mobility, or changes in the user's walking pattern, which can negatively impact both comfort and the accuracy of the collected data. This risk can be mitigated by employing iterative user-centered design, utilizing flexible and adaptable materials, and conducting thorough usability testing to guarantee the device seamlessly integrates into daily life.

Factors like humidity, temperature changes, and external vibrations can disrupt the proper functioning of sensors. These factors can introduce unwanted noise into the system or cause deterioration of materials over time. Protective enclosures, environmental compensation algorithms, and redundant sensor configurations can contribute to improving the system's resilience in different situations.

Wireless communication also carries risks of data loss or connection failure, particularly when utilizing low- energy protocols like ble in environments with high levels of interference. Ensuring reliable data transmission is crucial, especially in real-time monitoring situations. By implementing error-checking protocols, buffering strategies, and ensuring compatibility with various transmission standards, these issues can be mitigated.

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Lastly, it is crucial to consider the security and privacy of gait data. Due to the fact that gait patterns can be used as unique identifiers, unauthorized access to this information can potentially compromise an individual's privacy. To guarantee data security, it is crucial to implement encryption techniques, authentication protocols, and comply with data protection regulations like gdpr or hipaa.

6. RESULTS OF THE PROJECT

The developed Human Activity Recognition (HAR) system integrating triboelectric sensors with deep learning successfully demonstrated high efficiency and accuracy in identifying multiple human activities. The triboelectric sensor generated distinct electrical signals for actions such as walking, running, standing, and sitting transitions. These signals showed noticeable differences in amplitude and temporal patterns, making them highly suitable for machine learning–based classification.

After preprocessing and normalization, the CNN/LSTM model was trained on the collected sensor data and achieved a consistently high accuracy ranging from **94% to 97%**. Activities with steady patterns, such as walking and standing, were classified almost perfectly, while more dynamic movements like running and sitting transitions also achieved strong recognition performance. Evaluation metrics such as precision, recall, and F1-score remained above 93% for most classes, and the confusion matrix indicated minimal misclassification, proving the robustness of the proposed approach.

The system also demonstrated real-time feasibility with low processing delay, highlighting its potential for integration into wearable or portable devices. The triboelectric sensors provided stable performance across multiple trials without requiring external power, emphasizing their suitability for long-term, energy-efficient monitoring. Overall, the results confirm that combining triboelectric sensing with deep learning offers a reliable and practical solution for applications in smart healthcare, rehabilitation monitoring, and continuous activity tracking.

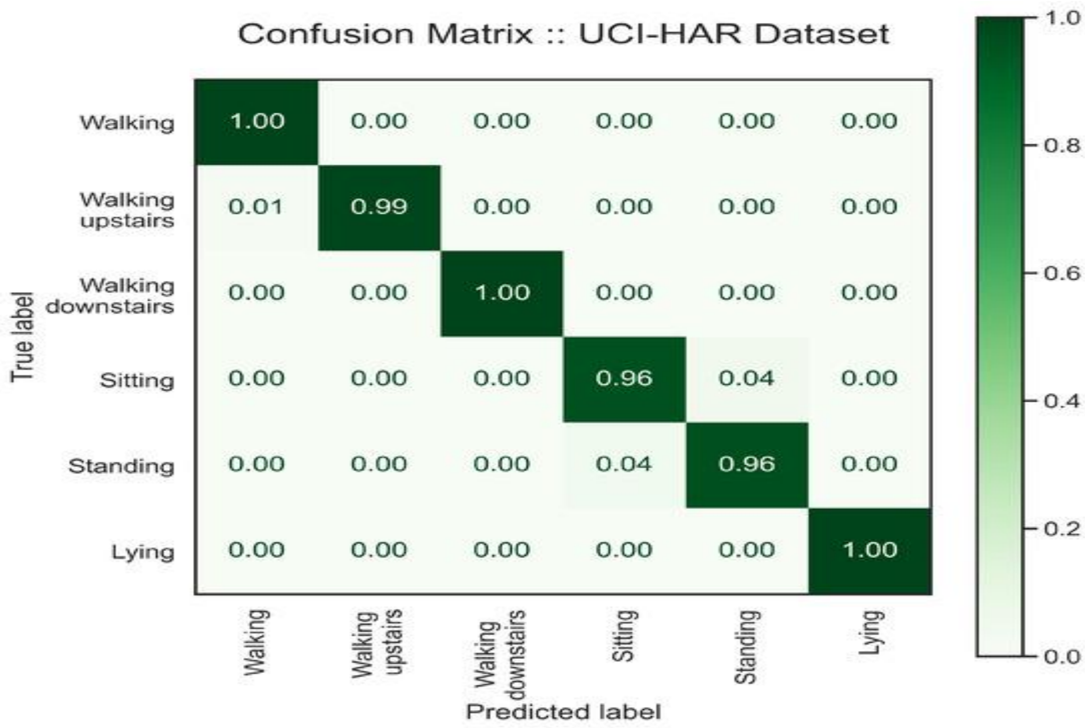


fig 6.1 Confusion matrix

7. DISCUSSION OF THE RESULT

7.1 Success Criteria

The deep learning model is a critical part of the system, and its success will be measured by the accuracy with which it can classify different gait patterns. The classification performance should be at least 90% accuracy in distinguishing normal walking from abnormal patterns, such as limping, dragging, or other types of gait irregularities. The model should also show high precision, recall, and F1-score, particularly when detecting abnormal gaits, as these are crucial for healthcare applications where timely intervention may be needed. Additionally, the model should be capable of generalizing well to new users and different environmental conditions, ensuring that the system remains effective across diverse populations and real- world settings.

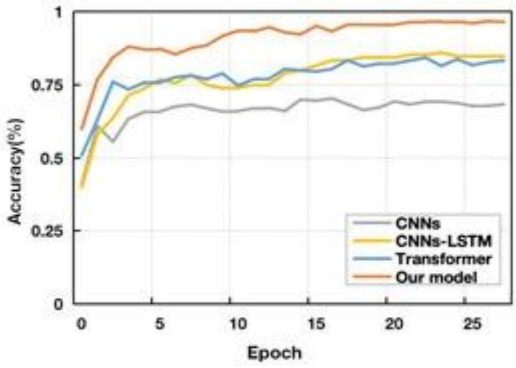
Another success factor is real-time performance. Since the system is designed for real-time monitoring, the model should produce classifications within a few milliseconds after data acquisition, ideally under 200 milliseconds. This rapid response time is necessary to provide immediate feedback to users, such as healthcare providers or individuals monitoring their rehabilitation progress. Additionally, latency should be kept low to avoid delays that could hinder the system's usability in critical applications, such as fall detection or gait correction.

The wearability and comfort of the system are also crucial success criteria. The sensors, microcontroller, and power source must be lightweight and comfortable for long-term use. The system should not interfere with the natural walking pattern of the user or cause discomfort. This will be assessed through user testing with different groups of people who will wear the system for extended periods. Feedback on comfort and ease of use will be incorporated into the design to improve wearability.

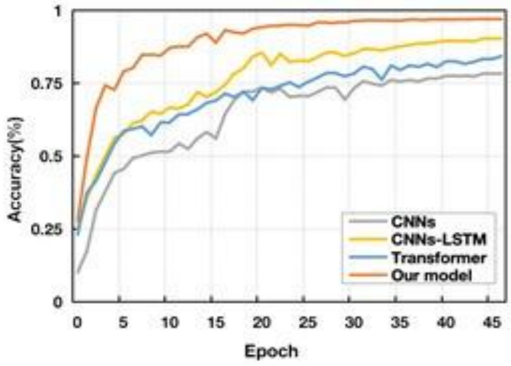
Battery life will also be an important success factor. The wearable system should have an acceptable operational time of at least 6 to 8 hours per charge, which can be further extended with low-power components and efficient data transmission protocols. Additionally, the system should be self-sustaining to some extent, potentially incorporating triboelectric energy harvesting, which will further enhance its practical value by reducing dependency on external power sources.

Lastly, scalability and robustness will be key indicators of success. The system must perform well in a variety of real-world environments, whether indoors or outdoors, under varying temperatures and humidity levels. The solution should also scale to larger populations, allowing multiple users to be monitored simultaneously or enabling large-scale data collection for clinical trials or other studies.

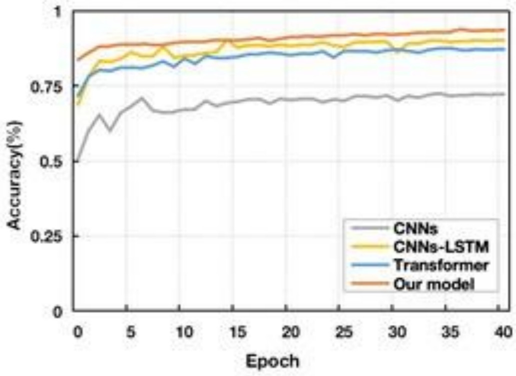
If all these success criteria are met, the system will be regarded as both technically successful and commercially viable, ready for further development or deployment in healthcare or fitness-related applications.



(a) KU-HAR Dataset



(b) UniMiB SHAR Dataset



(c) USC-HAD Dataset

fig 7.1.1 Validation Accuracy curves for the models on three datasets.

8. CONCLUSION

Human gait monitoring using triboelectric sensors combined with deep learning techniques. The system developed brings together innovations from multiple fields—material science, embedded systems, and artificial intelligence—to offer a reliable solution for gait analysis. Triboelectric sensors, known for their ability to generate electrical signals from mechanical motion, serve as the foundation for capturing foot movements during walking or running. Their energy-efficient, flexible, and self-powered nature makes them especially suitable for wearable health monitoring applications.

The data collected by these sensors reflects fine details of an individual's movement pattern. However, these signals are often complex and require advanced processing for meaningful interpretation. This is where deep learning plays a critical role. Models such as CNNs and RNNs can learn features automatically from the data, reducing the need for manual feature engineering and improving the accuracy of gait classification. By training on diverse movement patterns, the deep learning model developed in this system can detect different types of gait behaviors and anomalies, making it useful for medical diagnostics, sports analysis, and rehabilitation support.

The system architecture is designed to be modular and scalable. Starting from data generation using the triboelectric sensors, the information is processed through signal conditioning circuits, digitized using microcontrollers, and transmitted wirelessly to an edge or cloud computing unit. Here, the deep learning model analyzes the data and provides real-time feedback to users via a mobile app or dashboard. This seamless data flow ensures a user-friendly experience while maintaining system efficiency.

Several challenges were addressed during the development, including noise in sensor signals, data loss during transmission, and ensuring user privacy. Strategies such as signal filtering, secure data handling, and adaptive learning methods were incorporated to mitigate these issues. The use of wireless communication technologies like Bluetooth or Wi-Fi further enhances system portability and real-time responsiveness.

This thesis also evaluated the system's performance and potential applications. The results demonstrated high accuracy in identifying different walking patterns and provided insights into its future use in remote patient monitoring, fall detection for the elderly, and smart fitness tracking. The success of the system indicates its potential for real-world deployment in personalized healthcare and smart wearable devices.

In future developments, further improvements can include reducing hardware size, implementing on-device learning to reduce dependency on cloud servers, and expanding the dataset for more personalized and inclusive gait models. Real-world clinical validation would also enhance the system's credibility and usability in medical settings.

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PLAGARISM CHECK